Mo. Umer

Data Engineer | Analytics Engineer

abd.huss.work@gmail.com   +1 (631) 571-9172   Austin, TX, US

PROFESSIONAL SUMMARY

Data Engineer & Analytics Engineer with 8+ years experience in large-scale data pipelines, real-time ingestion systems, and semantic data models across energy, logistics, and media. Expert in end-to-end ELT pipelines on Snowflake, BigQuery, Databricks; hands-on with Apache Spark, Kafka, Apache Flink for real-time and streaming workloads. Specialized in CDC-based ingestion, dbt Semantic Layer, MetricFlow metric modeling, lakehouse architectures (Apache Iceberg, Delta Lake), Airflow orchestration across AWS, GCP, Azure. Strong in data governance, data contracts, data cataloging, schema governance, PII masking, RBAC, data observability; experienced with Docker, Kubernetes, Terraform, multi-cloud, CI/CD.s.

TECHNICAL SKILLS

Languages
Python, SQL, Scala, Bash

Data Warehouses and Lakehouses
Snowflake, BigQuery, Databricks, Apache Iceberg, Delta Lake, Amazon Redshift, Microsoft Fabric

Batch and Distributed Processing
Apache Spark, PySpark, Apache Hadoop, AWS EMR

Data Governance and Cataloging
Apache Atlas, OpenLineage, Marquez, data contracts, data mesh, RBAC, PII masking, column-level lineage

Data Quality and Observability
Monte Carlo, Great Expectations, dbt tests, Datafold

BI and Visualization
Looker, Tableau, Apache Superset, Power BI

Cloud Platforms
AWS (S3, Glue, EMR, Kinesis, Lambda, Glue Data Catalog), GCP (BigQuery, Dataflow, Pub/Sub), Azure (ADF, Synapse Analytics, Azure Data Factory)

Streaming and Messaging
Apache Kafka, Apache Flink, AWS Kinesis, Confluent Cloud, Debezium (CDC), Striim

Orchestration and Transformation
Apache Airflow, dbt (dbt Semantic Layer, MetricFlow), Prefect, Dagster

Databases
PostgreSQL, MySQL, MongoDB, Cassandra, Redis

DevOps and Infrastructure
Docker, Kubernetes, Terraform, GitHub Actions, Jenkins

PROFESSIONAL EXPERIENCE

Senior Data Engineer, *iStream Solutions* — 01/2022 – Present

- Architected a real-time event streaming platform using Apache Kafka and Apache Flink to process high-volume media consumption and user interaction events, delivering sub-second data availability across downstream data pipelines and analytical systems.

- Designed and implemented a lakehouse architecture on AWS using Apache Iceberg on S3, supporting time-travel queries, schema evolution, and partition pruning for large-scale historical data workloads without vendor lock-in.
- Built the organization's semantic layer using dbt Semantic Layer and MetricFlow to standardize business entity definitions and metric computation across product, marketing, and finance domains, enabling consistent, governed reporting across Looker and downstream BI tools.
- Engineered CDC-based data ingestion pipelines using Debezium and Confluent Cloud to replicate transactional database mutations into Snowflake with near-real-time data freshness, guaranteed ordering, and schema evolution support.
- Developed and maintained production Apache Airflow DAGs orchestrating multi-step ELT data pipelines, incorporating dependency management, SLA alerting, cross-system retry handling, and dynamic task generation for incremental data loads.
- Implemented a data observability framework using Monte Carlo to proactively detect freshness degradation, volume anomalies, and schema drift across data pipeline outputs, integrating alerts into on-call workflows.
- Established a data governance framework at iStream covering data contracts between producer and consumer teams, column-level lineage via OpenLineage, PII masking policies, RBAC access controls in Snowflake, and a centralized data catalog enabling self-service data discovery.
- Extended the multi-cloud data pipeline footprint by integrating Azure Data Factory pipelines for ingesting partner data feeds into the Snowflake lakehouse, ensuring consistent medallion architecture patterns across AWS and Azure workloads
- Containerized pipeline services with Docker and Kubernetes, managed infrastructure as code using Terraform, and integrated all data pipeline deployments into GitHub Actions CI/CD workflows for reliable, auditable releases.

Analytics Engineer, *KeepTruckin (Motive)* 07/2018 – 12/2021

- Built and maintained large-scale batch data pipelines using Apache Spark on AWS EMR to ingest and transform fleet telemetry, GPS event streams, and ELD compliance data feeding operational dashboards and downstream data modeling layers.
- Designed a multi-layer Snowflake data warehouse following the medallion architecture pattern, with clearly separated raw, cleansed, and aggregated layers serving diverse analytics, data science, and reporting consumers.
- Implemented dbt data models and applied data contract patterns to enforce schema agreements and ownership boundaries between producer pipelines and consuming analytics teams, reducing data quality incidents through versioned, tested, and documented transformations.
- Built streaming data pipelines using AWS Kinesis and Lambda functions to capture real-time vehicle events and persist them to S3-based data lakes in Parquet format with Hive-compatible partitioning and incremental load patterns.
- Migrated legacy cron-based data workflows to Apache Airflow DAG-driven orchestration, introducing dependency graphs, dynamic task generation, and integrated PagerDuty alerting, improving pipeline reliability and reducing manual intervention.
- Established schema registries for Kafka topics using Confluent Schema Registry, enforcing Avro-based data contracts and ensuring backward-compatible schema evolution across all streaming data pipelines.
- Automated data quality validation using Great Expectations at both ingestion and transformation stages, enforcing referential integrity, null rate thresholds, and value distribution constraints to maintain trusted data modeling outputs.

Data Engineer, *Enverus* 05/2017 – 06/2018

- Engineered ETL data pipelines to ingest, normalize, and integrate energy market data from structured and semi-structured third-party provider feeds into a centralized analytical environment on Amazon Redshift, applying data modeling best practices for dimensional schema design.
- Developed PySpark jobs on Hadoop clusters to process large-scale geospatial and well production datasets from oil and gas records, enabling complex analytical queries across multi-year historical partitions through optimized partition pruning strategies.
- Designed relational and dimensional data models supporting energy intelligence products, including well performance analysis datasets, production trend inputs, and regulatory compliance reporting structures used by external data subscribers.
- Built Apache Airflow orchestration workflows to automate daily and near-real-time data pipeline ingestion, transformation, and downstream delivery across multiple data domains and third-party API integrations.
- Implemented data lineage tracking using field-level provenance documentation and metadata management practices across all data pipelines to meet audit trail requirements for regulated energy market data, establishing early data governance standards for the platform.

SELECTED PROJECTS

Real-Time CDC Ingestion and Lakehouse Sync Pipeline

- Designed and deployed a change data capture data pipeline that captured row-level mutations from PostgreSQL transactional systems, streamed them through a Kafka-backed event bus with enforced Avro data contracts, and landed them in a Snowflake lakehouse with near-real-time latency, full column-level lineage via OpenLineage, and automated data quality gates at each stage.
- Tech Stack: Debezium, Apache Kafka, Confluent Cloud, Apache Flink, Snowflake, AWS S3, Apache Iceberg, OpenLineage, Terraform

dbt Semantic Layer and Analytics Engineering Platform with Data Governance

- Built a unified semantic layer on top of an Apache Iceberg-based lakehouse using dbt Semantic Layer and MetricFlow, establishing governed data contracts between producing data pipelines and consuming BI tools, implementing data catalog metadata via Apache Atlas, and creating a governed foundation for natural language query interfaces through well-defined data modeling and consistent business entity naming.
- Tech Stack: Apache Iceberg, dbt Semantic Layer, MetricFlow, Snowflake, AWS S3, Apache Airflow, Looker, Great Expectations, Datafold, Apache Atlas, OpenLineage

EDUCATION

Bachelor of Science in Computer Science